Task 2

Research Questions

1. What type of learning method best fits the purpose of this project? Why?

- Supervised learning is the best fit because in our case, we already give the model the samples with their designated label. All it has to do is classify the image according to its data.

2. a) Support Vector Classifiers - Vector drawn to divide datapoints to two sections. The vector is a hyperplane one dimension lower to the dimensions of the data

b) Support Vector Machine - It functions similar to the SVC. The difference is that it uses a Kernel function to scale the data to a higher dimension and draw a more accurate vector. It is most commonly used when many data you wish to classify one way is surrounded by data with the other type of classification and no SV would accurately divide them.

3. How do we train an SVM to classify our samples into healthy and unhealthy?

- Every label contains a mean NDVI value from 37 to 229. Since true NDVI classifications range from -1 to 1, we normalize our label values using the formula:

$$Y = 2*\left(\frac{x - min}{max - min}\right) - 1$$

- Anything under 0.5 will be classified as unhealthy and over will be healthy.

* How to evaluate models for accuracy, precision and recall?
- Create a table for predictions vs actual values

	Predicted Not	Predicted Healthy
Actual Not	True Negative	False Positive
Actual Healthy	False Negative	True Positive

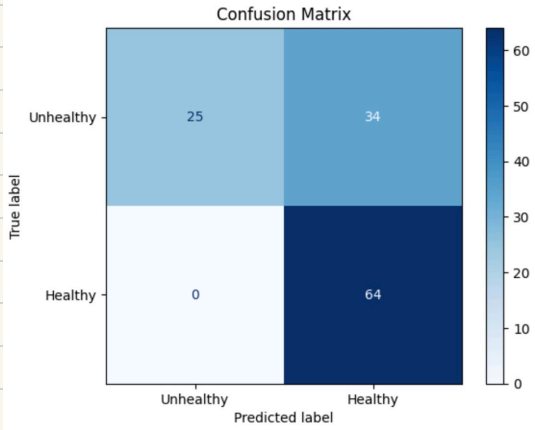
a) Accuracy - Measuses overall correctness
· Formula: $\frac{TP + TN}{\text{total images}}$ · sklearn: import accuracy_score

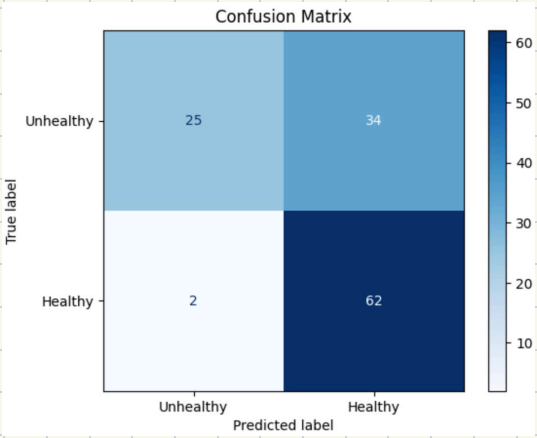
b) Precision - Measures how reliable positive predictions are
· Formula: $\frac{TP}{TP+FP}$ · sklearn: import precision_score

c) Recall - Measures how well the model detects actual positives!
· Formula: $\frac{TP}{TP+FN}$ · sklearn: import recall_score

* Linear Kernel scores:

Accuracy - 72.4%
Precision - 65.03%
Recall - 100%



* RBF Kernel scores:

Accuracy - 70.7%
Precision - 64.6%
Recall - 96.9%